CSE-476: Introduction to Natural Language Processing
Quiz 5

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

Q1. Which option best describes instruction tuning?

- (A) Pre-training on raw text with a next-word loss
- (B) Post-training on instruction–response pairs to follow tasks
- (C) Adding more layers to reduce perplexity values
- (D) Using masks to block attention to future tokens

Q2. Which benchmark is a widely used multiple-choice QA test for LLMs?

- (A) HotpotQA (generative multi-hop answers)
- (B) HumanEval (code generation via execution)
- (C) SQuAD (extractive span selection)
- (D) MMLU (multiple-choice benchmark)

Q3. A model answers 7 to ``A farmer has 10 sheep; all but 3 run away.'' What type of hallucination is this?

- (A) Factual hallucination (wrong world facts)
- (B) Commonsense hallucination (unsafe commonsense)
- (C) Reasoning hallucination (faulty logical steps)
- (D) No hallucination: answer is correct

Q4. Why use pairwise human preferences for alignment after instruction tuning?

- (A) They're easier than absolute "goodness" scores and capture relative quality
- (B) They strictly reduce compute and memory costs
- (C) They guarantee perfect annotator fairness
- (D) They remove the need for any supervised data

Q5. In PPO-based RLHF, what is the reward model's job?

- (A) Set attention masks during decoding
- (B) Collect human preference pairs online

(C) Tune optimizer hyper-parameters each step
(D) Score whole model responses so PPO can assign rewards

Q6. What is a key difference between DPO and PPO-based RLHF?

(A) DPO trains directly on preference pairs without a separate reward model
(B) DPO requires a value network to compute advantages
(C) DPO prohibits using preference data
(D) DPO only optimizes next-word likelihood

Q7. What primary issue is RAG introduced to help with?

(A) Grounding generations by retrieving external sources
(B) Increasing the parameter count to store more facts
(C) Replacing tokenization with sub-word merges
(D) Eliminating the need for instruction tuning

Q8. In the retrieval primer, what does BM25 add beyond plain TF-IDF?

(A) Backpropagates through retrieval to update encoders
(B) Byte-pair encoding to compress queries
(C) Learned neural embeddings for queries and documents
(D) Document-length normalization and tunable term saturation on term frequency

Q9. What does Recall@k measure?

(A) All retrieved docs
(B) Relevant docs retrieved out of all relevant ones
(C) Model accuracy
(D) Token overlap

Q10. In Dense Passage Retrieval (DPR), what are the Query Encoder and Passage Encoder typically based on?

(A) RNNs
(B) Decision Trees
(C) BERT-like models
(D) Linear Regression

End of Paper